

Conception, Development and First Evaluation of a Context-Adaptive User Interface for Commercial Vehicles

Lasse Schölkopf

Chair of Automotive Technology, Technical University of Munich

schoelkopf@ftm.mw.tum.de

Veronika Hutmann

MAN Truck & Bus SE, Munich

veronika.hutmann@man.eu

Maria-Magdalena Wolf

Chair of Automotive Technology, Technical University of Munich

maria.wolf@tum.de

Frank Diermeyer

Chair of Automotive Technology, Technical University of Munich

diermeyer@ftm.mw.tum.de

ABSTRACT

Modern vehicles are complex working environments and feature a multitude of functionalities. This applies in particular to commercial vehicles, which are equipped with even more functions than passenger cars. Research showed the potential of context-adaptive user interfaces for reducing the complexity of the human-machine interaction but has focused on passenger cars. Therefore, a context-adaptive touchscreen-based system is conceptualized specifically for commercial vehicles such as trucks based on existing findings and design guidelines. Acknowledging the importance of being able to gather early user feedback for allowing fast iteration cycles, an interactive prototype was implemented, and a modular study setup developed. This combination was tested in an initial user study, which evaluated the usability as well as user experience in terms of the novel context-adaptive interface.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods.

KEYWORDS

Context-Adaptive User Interface, Commercial Vehicles, User Study, Usability Evaluation, User Experience Evaluation

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1 INTRODUCTION

The task of driving can be described as a complex task [24]. At the same time, modern vehicles feature a high number of complex vehicle functions [25], which is steadily rising [11] and can distract the driver. Commercial vehicles such as trucks feature an even more complex driving workplace, partly due to a large number of highly specific control elements (e.g. for axle lifts) that are needed irregularly. Combined with the fact that large volumes of goods are transported across our roads by commercial vehicles [32] while most traffic accidents can be attributed to distractions from the driving task [8], the industry is in need of solutions to optimize human-machine interaction. Context-adaptive user interfaces (CAUI) have been identified as a promising solution for reducing complexity [17] and distractions from the driving task [35] in past research while improving user satisfaction [13]. CAUIs adapt to the current task or context [16] by changing the interface [1] and displaying the currently relevant functionalities. While such systems have been successfully explored in the past, the main focus was on personal cars and infotainment functionalities [3, 29, 33, 35, 36]. Furthermore, despite the advantages of CAUIs and their potential to simplify the human-machine interaction, challenges remain. CAUIs are inherently inconsistent over time [23], which can lead to unpredictable systems and subsequently disoriented users [9]. It can also be challenging to make the user understand the system behavior [35], accepting and trusting it [17, 35]. This could explain the relative unsuccessfulness of adaptive interfaces in the past [27] and should be addressed early in the design process, for example by providing the user with an appropriate mental model [26].

Therefore, this work aims to explore in an early development phase whether CAUIs are a suitable approach to optimize driving workplaces of commercial vehicles. A CAUI prototype was conceptualized, developed and evaluated to get initial user feedback. Furthermore, a modular technical setup allowing for rapid CAUI evaluations is presented.

2 USER INTERFACE CONCEPTION

The design phase built on already identified recurring situations during the daily work of truck drivers. Previously conducted user research revealed nine “working phases” such as *preparing for departure*, *empty drive*, *loading* as well as required vehicle functions for each phase [30]. These phases define the context of the CAUI,

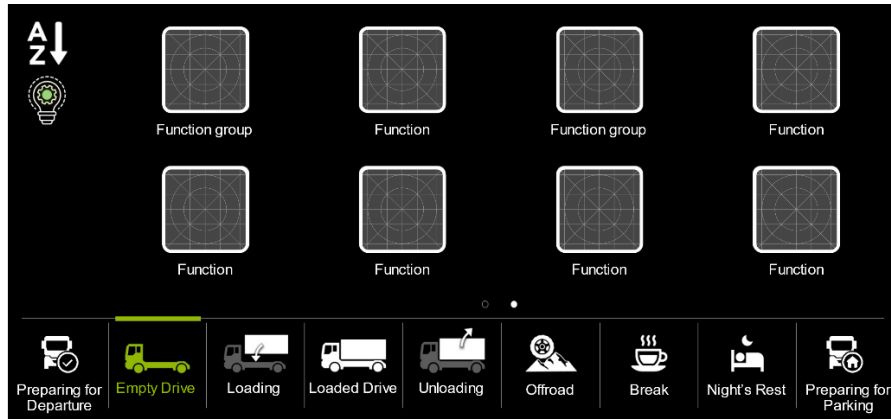


Figure 1: The conceptualized CAUI with the “Empty Drive” working phase selected, including schematic vehicle functions.

and the interface was structured around them. Similar vehicle functions were clustered by industry experts to form function groups, such as *traction* or *lighting*.

The created interface (Figure 1) highlights the currently active working phase, which is either automatically selected by the situation recognition or through user input. The vehicle functions contained in each working phase are ordered from high to medium relevancy. For increased user control, the situation recognition can be deactivated with the light bulb icon at the top left and all functions can be shown in alphabetical order via the A-Z icon. To achieve high usability and user experience, the design is based on regulations for automotive UIs and guidelines for interactive interfaces: Font sizes were designed to surpass 16 arc minutes to guarantee “acceptable” readability [20] and the minimal contrast between text and background exceeds a ratio of 1:5 required by ISO 15008 [20]. The maximum number of visible items (i.e., function group or function) was limited to 8 to minimize short-term memory load [31]. For working phases requiring more than 8 functions horizontally swiping pages with page indicators at the bottom were implemented. Interaction elements as well as the layout were adopted from Google’s Material Design guidelines [14] and ISO EN ISO 9241-125 [10]. Moreover, a truck-specific iconography was designed.

Next, we turned the concept into an interactive prototype to enable user testing early in the development process and therefore reducing costly late changes [12]. Early prototyping allows for an iterative design process through continuous user feedback [21], which helps making critical improvements to the user interface [18].

3 IMPLEMENTATION AND TEST SETUP

The CAUI prototype was implemented on a 10.1-inch touchscreen (1540x720 pixels) which features a force sensor as well as a haptic feedback actuator (Figure 3, b). The user applies a predefined force to the screen in order to activate a function and receives haptic feedback which feels like pressing a mechanical button, whereas a light touch only highlights the button (Figure 2). It is known that haptic feedback reduces gaze from the road [5] as well as improves

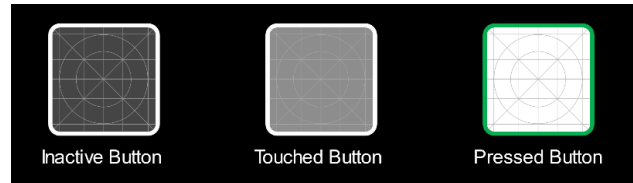


Figure 2: Implemented button states.

the user experience [15, 28] and potentially facilitates operation in rough road conditions.

The prototype was integrated in a static truck driving simulator at the Technical University of Munich (Figure 3, a) consisting of three 55-inch LCD Monitors (3840x2160 pixels) that enable manual driving tasks or playback of driving videos. The modularity of the test setup allows the CAUI to be controlled via either situational recognition algorithms or a human operator. The latter setup is known as Wizard-of-Oz and is especially useful in early phases of development [2]. In this work, the Wizard mimicked the algorithm and changed the working phase by pressing predefined keys on a keyboard. The prototype was implemented in Unity [34] and loads all items dynamically, allowing a relocation of all control elements during runtime.

This could be used in future versions to position frequently used items over time at the top of the grid. Additionally, the prototype can be extended to send signals via CAN bus to trigger actions in the simulation environment or physical control modules in future studies.

4 USER STUDY

A pilot user study was conducted to obtain initial user feedback on the CAUI, investigate usability issues and establishing baselines regarding usability and user experience scores. Furthermore, subjective impressions of the test subjects were of interest to improve the interface. Moreover, we wanted to investigate whether the test setup and the prototype are suitable for conducting user studies.

While it is advised to evaluate new products by representative users [6, 18], the participant population was limited to a university

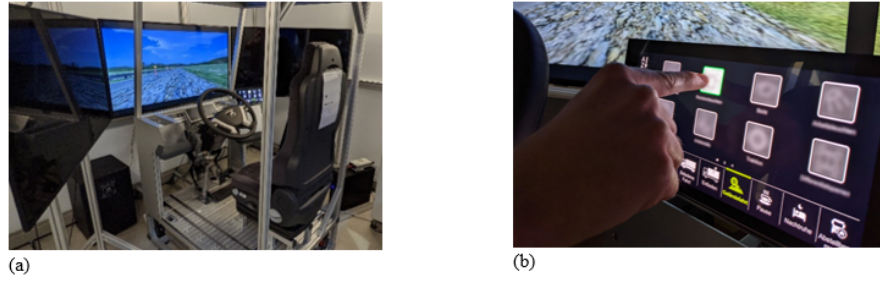


Figure 3: Static truck simulator with integrated prototype (a), haptic touchscreen with implemented CAUI (b)

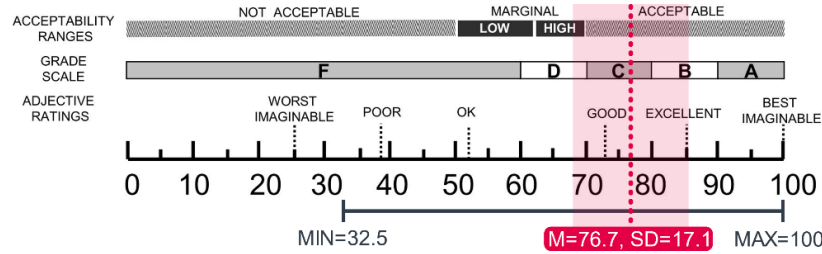


Figure 4: SUS Score of 76.7 (dashed line) with standard deviation (transparent overlay) and score range from 32.5 to 100 according to [4].

environment due to the coronavirus pandemic. The volunteering sample $N=35$ consisted of 7 females and 28 males, all of whom held a driver's license and were unfamiliar with the system. The average age of the participants was 28.9 years ($SD=11.3$) while they drove on average 8257.9 kilometers per year ($SD=7215.4$).

The study was conducted under strict adherence to a hygiene concept and took one hour per participant. After collecting demographic data, the participant was seated in the simulator and explored the system for up to 5 minutes. Next, the user was asked to imagine being a driver for a construction company and having been assigned to pick up debris from a construction site. He was asked to activate vehicle functions when instructed to do so by the investigator via an intercom system. During the run, the driving simulation played back a prerecorded video of a truck driver's typical working day, which contained 8 different work phases with a total of 11 phase changes, since some occurred more than once. The subject was asked to activate or deactivate various vehicle functions, accounting for a total of 30 interactions during the 15-minute drive. After completion, a survey containing standardized questionnaires was filled out. The CAUI's usability was assessed by the System Usability Scale (SUS) [7] questionnaire because of its fast execution and understandability [18] while the user experience was measured via the User Experience Questionnaire (UEQ) [19, 22]. The traceability of the CAUI's actions was rated via the question, "Did you find the behavior of the system traceable?" on a 7-point Likert scale ("not traceable at all" to "fully traceable"). To obtain qualitative feedback, positive and negative characteristics of the system were asked in open questions.

5 RESULTS

All collected quantitative data was anonymized and statistically analyzed. The participants rated the CAUI's usability on average with a SUS score of 76.7 ($SD=17.1$), putting it well within the "acceptable" range [4] and surpassing the adjective rating of "good" (Figure 4).

For the user experience, three participants were excluded due to inconsistencies [19]. The UEQ's values lie for each dimension (Figure 5) in the green and therefore acceptable range. The CAUI's attractiveness was rated on average at 1.78 ($SD=0.83$), whilst the pragmatic qualities of the system scored lower at 1.61 ($SD=1.19$) for perspicuity, 1.43 ($SD=1.12$) for efficiency and 1.43 ($SD=1.24$) for dependability. The hedonic dimensions achieved higher ratings of 1.63 ($SD=0.91$) for stimulation and 1.55 ($SD=0.88$) for novelty.

The traceability reached a score of 6.0 ($SD=1.5$) on a scale from 1 to 7, indicating a good comprehensibility of the system's actions. The qualitative data resulting from the open questions was analyzed by clustering similar statements. The following are the three most frequently mentioned positive and negative comments: Good haptic feedback of the touchscreen (17 mentions), high intuitiveness and ease of use after a short training period (13 mentions), good menu navigation and structure (12 mentions). As the top point of criticism, the categorization of functions needs improvement (17 mentions) while the changing order of functions per working phase confuses (11 mentions) and the depth and width of the menu is too complex (10 mentions). With regard to the suitability of the implementation and the test setup, only minor and isolated errors were found during the course of the study.

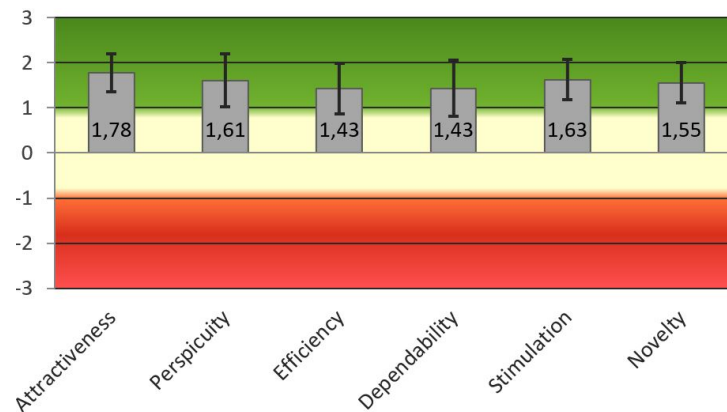


Figure 5: UEQ Score for each of the questionnaire's dimension with mean values and standard deviation according to [19].

6 DISCUSSION AND CONCLUSION

In this work, an initial CAUI concept for trucks as well as a dynamic and expandable prototype implementation was presented. Furthermore, a test setup allowing for fast iterations and evaluations of the system was shown and proven to work in a pilot user study. Only minor errors occurred, which are assumed to have no significant impact on the results.

The CAUI's usability and user experience were evaluated and showed encouraging results for a first version. The usability score lies within the acceptable range and is supported by the high perceived traceability of the system's actions. Hedonic dimensions of the user experience were rated higher than pragmatic qualities, indicating that future iterations should focus on improving the effectiveness, dependability and perspicuity. First points of improvement can be derived from the qualitative feedback such as the criticism of the categorization of vehicle functions. However, we hypothesize that these negative comments can be attributed to the non-representativeness of the sample group, since truck drivers would have been more familiar with the specific functionalities presented by the CAUI. Therefore, a validation of the categorization with truck drivers in the future is advised. Carrying out future studies with a representative sample group also would increase the validity and transferability of the results. Nevertheless, recruiting subjects from a university environment allowed for a first rapid evaluation despite the pandemic situation and identified initial usability improvements. In addition, future studies should involve an active driving task in order to investigate the suitability of the CAUI for interactions while manually driving a vehicle.

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He revised the paper critically for important intellectual content. Frank Diermeyer gave final approval of the version to be published and agrees to all aspects of the work. As a guarantor, he accepts responsibility for the overall integrity of the paper. This research was funded by MAN Truck & Bus SE.

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